

AURA V5 — COMPLETE DATA INTELLIGENCE REPORT

Property Intelligence Infrastructure — Data Layer Definition & Signals

1 INTRODUCTION

AURA V5 is not a traditional machine learning pipeline. It is a **multi-layer real estate intelligence infrastructure** designed to:

- Estimate **fair property value**
- Explain **why value exists**
- Detect **hidden opportunities**
- Enable **seller-first strategy**

The system is built on **modular data layers (Categories)**, each representing a distinct type of intelligence.

2 GLOBAL ARCHITECTURE

AURA V5 is structured into:

CORE → TRUTH → MARKET → CONTEXT → SPATIAL → BEHAVIOR → DESIRABILITY

Each layer contributes a different dimension of understanding.

3 CATEGORY DEFINITIONS & SIGNALS

CAT1 — PROPERTY INTELLIGENCE (CORE)

Role

Represents the **intrinsic characteristics of a property**

Level

Property-level

Key Signals

◆ Area Intelligence

- `area_effective` → usable surface
- `log_area` → normalized size

👉 Captures **scale effect on price**

◆ Room Intelligence

- `rooms`, `bedrooms`, `bathrooms`
- `room_density`
- `bathroom_density`

👉 Measures **space efficiency**

◆ Parking Signal

- `parking_score`

👉 Adds **premium value indicator**

◆ Age Intelligence

- `property_age`

- `age_score`

👉 Reflects **depreciation vs modernity**

◆ Property Encoding

- `property_type_enc`
- `energy_efficiency_level_enc`

👉 Enables modeling of **categorical differences**

🔥 Composite Signal

- `property_score`

👉 Global property attractiveness indicator

🟠 CAT1.2 — MARKET CALIBRATION

🎯 Role

Bridges **property** → **market pricing logic**

Signals

- price normalization
- calibration weights
- price scaling

👉 Ensures model aligns with **market reality**

🟡 CAT2 — MARKET / HPI (LOCAL PRICE DYNAMICS)

🎯 Role

Captures **market evolution over time**

Level

Municipality / regional

Signals

- price trends
- market growth
- volatility

👉 Defines:

👉 **Local market truth**

CAT3 — MACROECONOMIC CONTEXT

Role

Captures **external economic forces**

Signals

- interest rates
- inflation
- financial constraints

👉 Defines:

👉 **affordability ceiling**

CAT4 — SOCIOECONOMIC LAYER

Role

Captures **population and social structure**

Signals

- income levels
- demographics
- population density

👉 Defines:

👉 **demand potential**

CAT5 — GEO / URBAN INTELLIGENCE (**CORE MOAT**)

Role

Transforms building-level data into **urban structure intelligence**

Level

Grid (aggregated from ~2M buildings)

SIGNAL GROUPS

◆ **Geometry Intelligence**

- compactness
- elongation_ratio
- coverage_ratio
- shape_complexity

👉 Captures **building form & efficiency**

◆ **Density Intelligence (🔥 CRITICAL)**

- buildings_in_grid
- local_density_score
- density_index
- urban_pressure_score

👉 Measures:

👉 **urban intensity & demand concentration**

◆ **Morphology Intelligence**

- morphology_score
- mean_compactness

👉 Captures:

👉 **neighborhood consistency**

◆ **ESG Intelligence**

- energy_score_clean
- renovation_score_clean
- sustainability_index

👉 Reflects:

👉 **efficiency + upgrade potential**

◆ **Geo Anchors**

- centroid_lat / lon
- grid_id_250m

👉 Enables:

👉 **fusion across datasets**

CAT6 — BEHAVIORAL INTELLIGENCE

Role

Captures **market dynamics from listings**

Signals

- price changes
- volatility
- listing behavior

👉 Defines:

👉 **seller behavior & urgency**

CAT7 — TRANSACTION TRUTH (🔥) MOST IMPORTANT

Role

Represents **real market pricing reality**

Level

Municipality + time

SIGNALS

- ◆ **Core Price**

- `price_per_m2`

👉 Real valuation anchor

♦ Temporal Signals

- `year`
 - `quarter`
-

♦ Market Dynamics

- `price_growth`
- `price_trend`
- `price_volatility`

👉 Captures:

👉 **true market evolution**

🚨 Importance

Without Cat7:

✗ AURA = listing-based model

With Cat7:

✓ AURA = **reality-based system**

🟦 CAT8 — POI / ACCESSIBILITY (DESIRABILITY ENGINE)

🎯 Role

Measures **how attractive a location is**

Level

Grid-level

SIGNALS

◆ Distance Features

- distance to POIs
 - nearest service proximity
-

◆ Density Features

- POI density (1km)
 - category densities
-

◆ Accessibility Scores

- transport_score
 - healthcare_score
 - education_score
 - retail_score
-

Composite Signal

- `accessibility_score`

 Represents:

15-minute city logic

DATA FLOW

```
RAW DATA
↓
FEATURE ENGINEERING
↓
CATEGORY LAYERS
↓
AGGREGATION (Cat5)
↓
ALIGNMENT
↓
FUSION (NEXT STEP)
↓
AURA CORE DATASET
```

5 KEY DESIGN PRINCIPLES

◆ 1. Layered Intelligence

Each category = different signal type

→ avoids feature mixing

◆ 2. No Leakage

Removed:

- derived labels
 - target-like variables
-

◆ 3. Explainability

Each signal is interpretable:

- density → demand
 - age → depreciation
 - accessibility → desirability
-

◆ 4. Data Moat Strategy

Cat5 + Cat8 create:

👉 **defensible proprietary intelligence**

6 CURRENT SYSTEM STATUS

Category	Status
Cat1	✓ Production ready
Cat1.2	✓ Ready
Cat2	⚠ Needs localization upgrade
Cat3	✓ Stable
Cat4	✓ Stable
Cat5	🔥 Strong (aggregated)
Cat6	✓ Good
Cat7	🔥 Critical — ready
Cat8	⚠ Usable (upgrade later)

7 LIMITATIONS

- No unified dataset yet
 - Cat8 not fully optimized
 - Cat2 still partially aggregated
 - No feedback loop (Cat9)
-

8 STRATEGIC VALUE

AURA V5 enables:

- explainable valuation

- seller-first targeting
 - opportunity detection
 - market intelligence
-

FINAL CONCLUSION

AURA V5 data system is:

- ✓ Structurally complete
- ✓ Rich in signals
- ✓ Ready for fusion

But:

 **Value is not unlocked until fusion**

NEXT STEP

AURA V5 — FUSION PHASE

This will:

- align all categories
 - create unified dataset
 - enable modeling
-

FINAL TAKE

You have successfully built:

| one of the hardest parts of real estate AI systems

Now:

👉 everything must connect.

If you want next:

👉 I turn this into a **PDF / report format with diagrams + visuals**

OR

👉 we go straight into

FUSION STEP 1